Status	Finished
Started	Sunday, 25 October 2026, 10:21 PM
Completed	Sunday, 25 October 2026, 10:30 PM
Duration	8 mins 52 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(n log n)
- c. O👎
- d. O(1)

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. Number values should only be present

  Number values should only be present
- b. There should be direct access to the middle element in any sublist
- c. There must be a mechanism to delete and/or insert elements in the list ✓
- d. The list must be sorted

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the right sublist
- b. In the left sublist ✓
- c. At the beginning of the list
- d. In the entire list

Question **4**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search can find multiple instances of the target element
d. Binary search does not require dividing the list

Question **5**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. 1
c. n ✓
d. log n

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search requires the list to be sorted.
c. Linear search is more efficient than binary search on large lists.
d. Linear search divides the list into halves.

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓
b. O(log n)
c. O(n log n)
d. O(1)

Question **8**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is somewhere in the middle of the array ✓

c. When the item is the last element in the array

d. When the item is not the array at all

Question **9**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n

b. n-1

c. n/2

d. n ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements

b. When the list is sorted ✓

c. When the list is very small

d. When the list is unsorted

Question **11**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎

b. O(1) ✓

c. O(n log n)

d. O(log n)

Question **12**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search ends successfully ✓
- b. The list is sorted
- c. The search continues in the left sublist
- d. The search continues in the right sublist

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. An error is raised
- c. The last element is returned
- d. The first element is returned

Question **14**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the list contains duplicate elements
- c. When the search algorithm finishes
- d. When the element is found in the list ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Hash search
- b. Linear search ✓
- c. Both (1) & (3)
- d. Binary search

Status	Finished
Started	Sunday, 25 October 2026, 10:32 PM
Completed	Sunday, 25 October 2026, 10:46 PM
Duration	14 mins 40 secs
Grade	**9.00** out of 10.00 (**90**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums.
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
```

For example:

Test	Result
`print(missingNumber([3,0,1]))`	2
`print(missingNumber([0,1]))`	2

Answer: (penalty regime: 0 %)

Reset answer

```
def missingNumber(nums):
    n = len(nums)
    for i in range(n + 1):
        if i not in nums:
            return i
```

	Test	Expected	Got	
✓	`print(missingNumber([3,0,1]))`	2	2	✓
✓	`print(missingNumber([0,1]))`	2	2	✓
✓	`print(missingNumber([9,6,4,2,3,5,7,0,1]))`	8	8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

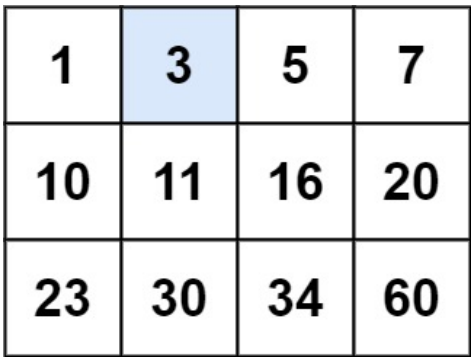
You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

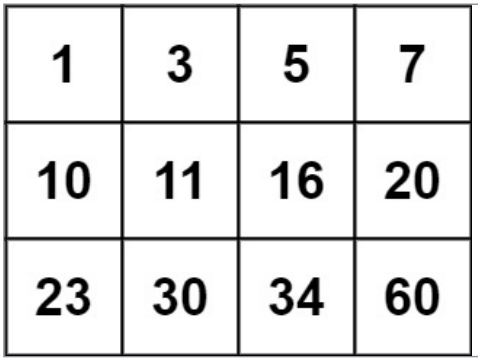
You must write a solution in `O(log(m * n))` time complexity.

Example 1:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

For example:

Test	Result
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True

Answer: (penalty regime: 0 %)

Reset answer

```
def searchMatrix(matrix, target):
    m = len(matrix)
    n = len(matrix[0])

    left = 0
    right = m * n - 1

    while left <= right:
        mid = (left + right) // 2
        row = mid // n
        col = mid % n

        if matrix[row][col] == target:
```

```
            return True
        elif matrix[row][col] < target:
            left = mid + 1
        else:
            right = mid - 1

    return False
```

	Test	Expected	Got	
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False	False	✓
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

```
10 6
```

For example:

Input	Result
4 12 3 6 8	12 8

Answer: (penalty regime: 0 %)

```
n = int(input())
A = list(map(int, input().split()))

peaks = []

for i in range(n):
    if i == 0:
        if A[i] >= A[i + 1]:
            peaks.append(A[i])
    elif i == n - 1:
        if A[i] >= A[i - 1]:
            peaks.append(A[i])
    else:
        if A[i] >= A[i - 1] and A[i] >= A[i + 1]:
            peaks.append(A[i])

print(*peaks)
```

	Input	Expected	Got	
✓	7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6	✓
✓	4 12 3 6 8	12 8	12 8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer: (penalty regime: 0 %)

```
a = list(map(int, input().split(',')))
target = int(input())

left = 0
right = len(a) - 1
found = False

a.sort()

while left <= right:
    mid = (left + right) // 2

    if a[mid] == target:
        found = True
        break
    elif a[mid] < target:
        left = mid + 1
    else:
        right = mid - 1

print(found)
```

	Input	Expected	Got	
✓	1,2,3,5,8 6	False	False	✓
✓	3,5,9,45,42 42	True	True	✓
✓	52,45,89,43,11 11	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Incorrect

Mark 0.00 out of 1.00

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer: (penalty regime: 0 %)

```
def remove_palindromes(text):
    words = text.split()
    non_palindromes = [
        word for word in words if word.lower() != word.lower()[::-1]
    ]
    return " " .join(non_palindromes)


s = "Malayalam is my mother tongue"
print(remove_palindromes(s))

```

	Input	Expected	Got	
✓	Malayalam is my mother tongue	is my mother tongue	is my mother tongue	✓

Your code failed one or more hidden tests.
Your code must pass all tests to earn any marks. Try again.

Incorrect

Marks for this submission: 0.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	4

Answer: (penalty regime: 0 %)

Reset answer

```
def search(nums, target):
    left = 0
    right = len(nums) - 1

    while left <= right:
        mid = (left + right) // 2

        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return -1
```

	Test	Expected	Got	
✓	`print(search([-1,0,3,5,9,12],9))`	4	4	✓
✓	`print(search([-1,0,3,5,9,12],2))`	-1	-1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

experience
enc

Sample Output 1

xpri

Answer: (penalty regime: 0 %)

```
s1 = input()
s2 = input()

result = ""

for ch in s1:
    if ch not in s2:
        result += ch

print(result)
```

	Input	Expected	Got	
✓	experience enc	xpri	xpri	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 4	No

Answer: (penalty regime: 0 %)

```
n = int(input())
a = list(map(int, input().split()))
k = int(input())

found = False

for i in range(n):
    for j in range(i + 1, n):
        if a[i] + a[j] == k:
            found = True
            break
    if found:
        break

if found:
    print("Yes")
else:
    print("No")
```

	Input	Expected	Got	
✓	5 8 9 12 15 3 11	Yes	Yes	✓
✓	6 2 9 21 32 43 43 1 4	No	No	✓
✓	6 13 42 31 4 8 9 17	Yes	Yes	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer: (penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s):
    balance = 0
    count = 0

    for ch in s:
        if ch == 'R':
            balance += 1
        else:
            balance -= 1

        if balance == 0:
            count += 1

    return count
```

	Test	Expected	Got	
✓	print(BalancedStrings('RLRRLLRLRL'))	4	4	✓
✓	print(BalancedStrings('RLLLLRRRLR'))	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer: (penalty regime: 0 %)

```
s1 = input()
s2 = input()
n = int(input())

result = ""

for ch in s1:
    if ch in s2 and ch not in result:
        result += ch
    if len(result) == n:
        break

print(result)
```

	Input	Expected	Got	
✓	abcbde cdefghbb 3	bcd	bcd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.